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POWERED FASTENER DRIVER

Abstract

A powered fastener driver includes a housing, a cylinder within the housing, the cylinder containing a pressurized gas, a piston within the cylinder and movable from a top-dead-center position to a bottom-dead-center position, a driver blade movably coupled to the piston, a lifter, a drive unit configured to provide torque to the lifter, and a sensor. The driver blade has a plurality of teeth. The sensor is configured to detect one or more of the plurality of teeth and a controller in communication with the drive unit and the sensor. The controller is configured to count a number of the plurality of teeth detected by the sensor, compare the number of detected teeth to a predetermined number of teeth, and stop the drive unit when the number of detected teeth is less than the predetermined number of teeth.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of co-pending U.S. patent application Ser. No. 18/422,512 filed on Jan. 25, 2024, which claims priority to U.S. Provisional Patent Application No. 63/591,844 filed on Oct. 20, 2023, and U.S. Provisional Patent Application No. 63/481,929 filed on Jan. 27, 2023, the entire contents of all of which are incorporated herein by reference.

FIELD OF THE INVENTION

[0002] The present disclosure relates to power tools, and more specifically to powered fastener drivers.

BACKGROUND OF THE INVENTION

[0003] There are various fastener drivers used to drive fasteners (e.g., nails, tacks, staples, etc.) into a workpiece known in the art. These fastener drivers operate utilizing various means (e.g., compressed air generated by an air compressor, gas spring, or the like) known in the art. Over the lifetime of the fastener driver, components of the tool may wear, which can cause the driver to fail.

SUMMARY OF THE INVENTION

[0004] In some aspects, the techniques described herein relate to a powered fastener driver including: a housing; a cylinder within the housing, the cylinder containing a pressurized gas; a piston within the cylinder and moveable from a top-dead-center position to a bottom-dead-center position; a driver blade moveably coupled to the piston for driving a fastener into a workpiece, the driver blade having a plurality of teeth; a lifter configured to engage the plurality of teeth for returning the driver blade from the bottom-dead-center position toward the top-dead-center position; a drive unit configured to provide torque to the lifter, causing it to rotate; a sensor configured to detect one or more of the plurality of teeth; and a controller in communication with the drive unit and the sensor, the controller configured to: count a number of plurality of teeth detected by the sensor, compare the number of detected teeth to a predetermined number of teeth, and stop the drive unit when the number of detected teeth is less than the predetermined number of teeth.

[0005] In some aspects, the techniques described herein relate to a powered fastener driver including: a housing; a cylinder within the housing, the cylinder containing a pressurized gas; a piston within the cylinder and moveable from a top-dead-center position to a bottom-dead-center position; a driver blade moveably coupled to the piston for driving a fastener into a workpiece, the driver blade having a plurality of teeth; a frame supported within the housing; a lifter supported within the housing by the frame and configured to engage the plurality of teeth for returning the driver blade from the bottom-dead-center position toward the top-dead-center position; a drive unit configured to provide torque to the lifter, causing it to rotate; a lifter position sensing assembly configured to detect an angular position of the lifter at an end of a fastener driving cycle, the lifter position sensing assembly including a sensor coupled to the frame and a sensor target coupled for

co-rotation with the lifter; and a controller in communication with the drive unit and the lifter position sensing assembly, the controller configured to stop the drive unit in response to the lifter position sensing assembly not detecting the angular position of the lifter at the end of the fastener driving cycle. Features and aspects of the disclosure will become apparent by consideration of the following detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is perspective view of a powered fastener driver.

[0007] FIG. 2 is another perspective view of the powered fastener driver of FIG. 1, with portions of a housing removed.

[0008] FIG. 3 is a partial cross-sectional view of the gas spring-powered fastener driver of FIG. 1.

[0009] FIG. 4 is another partial cross-sectional view of the gas spring-powered fastener driver of FIG. 1, illustrating a sensor proximate a driver blade.

[0010] FIG. 5 is a schematic view of a control system of the powered fastener driver of FIG. 1.

[0011] FIG. 6 is a graph illustrating a position of the driver blade during a fire-reload cycle.

[0012] FIG. 7 is a graph illustrating a simulated jam condition of the driver blade.

[0013] FIG. 8 is perspective cross-sectional view of a gas spring-powered fastener driver in accordance with another embodiment of the invention.

[0014] FIG. 9 is a partial cross-sectional view of the gas spring-powered fastener driver of FIG. 8.

[0015] FIG. 10 is a side view of a portion of the gas spring-powered fastener driver of FIG. 8 with portions removed to illustrate a lifter assembly.

[0016] FIG. 11 is a top view of a portion of the gas spring-powered fastener driver of FIG. 8, illustrating a lifter position sensor assembly coupled to the lifter assembly.

[0017] FIG. 12 is a schematic view of a control system of the powered fastener driver of FIG. 8.

[0018] Before any embodiments of the disclosure are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

DETAILED DESCRIPTION

[0019] With reference to FIGS. 1-3, a gas spring-powered fastener driver 10 is operable to drive fasteners (e.g., nails, tacks, staples, etc.) held within a magazine 14 into a workpiece. The fastener driver 10 includes a cylinder 18 containing a pressurized gas. A piston 22 is supported within the cylinder 18 and is moveable from a top-dead-center position to a bottom-dead-center position. The cylinder 18 includes a drive cylinder 25 having an open end in fluid communication with a storage chamber cylinder 27, which exposes the piston 22 positioned within the drive cylinder 25 to the pressurized gas in the storage chamber 27. The fastener driver 10 further includes a driver blade 26 that is attached to the piston 22 and moveable therewith.

[0020] The fastener driver 10 includes a housing 30 having a cylinder housing portion 34 and a motor housing portion 38 extending therefrom. The cylinder housing portion 34 is configured to support the cylinder 18, whereas the motor housing portion 38 is configured to support a drive unit 40. The drive unit 40 includes an electric motor 42 and a transmission 82 positioned downstream of the motor 42. In addition, the illustrated housing 30 includes a handle portion 46 extending from the cylinder housing portion 34, and a battery attachment portion 50 coupled to an opposite end of the handle portion 46. A battery pack 54 is removably coupled to the battery attachment portion 50 and supplies electrical power to the drive unit 40. The handle portion 46 supports a trigger 58,

which is depressed by a user to initiate a driving cycle of the fastener driver **10**.

[0021] With reference to FIG. 3, the driver blade **26** defines a driving axis **62** and includes a plurality of driver blade teeth or lift teeth **74** formed along an edge **78** of the driver blade **26**, which extends in the direction of the driving axis **62**. In particular, the lift teeth **74** project laterally from the edge **78** relative to the driving axis **62**. During a driving cycle, the driver blade **26** and piston **22** are moveable along the driving axis **62** between a top-dead-center (TDC) position and a bottom-dead-center (BDC) or driven position. The driver blade **26** may further be held in a ready position, which is positioned between the BDC position and the TDC position. The piston **22** is adjacent a top end **19** (FIG. 3) of the cylinder **18** in the TDC position, and the piston **22** is adjacent a bottom end **20** (FIG. 3) of the cylinder **18** in the BDC position. The fastener driver **10** further includes a rotary lifter **66** supported within the housing **30** by a frame **70** (FIG. 2). The rotary lifter **66** includes a plurality of rollers **90** supported by a plurality of pins **94**. The lifter **66** is supported on a lifter frame **70** and receives torque from the drive unit **40**, causing the lifter **66** to rotate. The lifter **66** and the drive unit **40** may be collectively referred to as a lifter assembly **88** (FIG. 2). As the lifter **66** rotates, the rollers **90** sequentially engage the lift teeth **74** formed on the driver blade **26** to return the driver blade **26** along the driving axis **62** from the BDC position toward the TDC position.

[0022] The cylinder **18** includes a bumper **98** located at the bottom end **20** of the cylinder **18**. The bumper **98** has a generally annular, frusto-conical shape with a central bore **99** therethrough. The bore **99** is coaxial with the driving axis **62** such that the driver blade **26** extends through the bore **99**. As the piston **22** and the driver blade **26** move from the TDC position toward the BDC position, the piston **22** impacts the bumper **98**, which absorbs the impact from the piston **22** and stops the piston **22** in the BDC position. In some embodiments, the bumper **98** is constructed of a resilient material (e.g., rubber, elastomeric material, or the like).

[0023] As illustrated in FIG. 5, the fastener driver **10** includes a jam detection system **100** having a controller **110**, one or more sensors **101** in communication with the controller **110**, and an indicator **112** in communication with the controller **110**. The controller **110** includes a plurality of electrical and electronic components that provide power, operational control, and protection to the components and modules within the controller **110** and/or the nailer **10**. The indicator **112** may be exposed from an exterior of the housing **30** and may be configured as one or more of lights (e.g., a light-emitting diode or LED), a display panel, or the like. The controller **110** may selectively activate the indicator **112** to alert the operator that maintenance is required.

[0024] For example, the controller **110** may include, among other things, a processing unit **124** (e.g., a microprocessor, a microcontroller, or another suitable programmable device) and a memory **128**. The memory **128** is a non-transitory computer readable medium and includes, for example, a program storage area and a data storage area. The program storage area and the data storage area can include combinations of different types of memory, such as a ROM, a RAM (e.g., DRAM, SDRAM, etc.), EEPROM, flash memory, a hard disk, an SD card, or other suitable magnetic, optical, physical, or electronic memory devices. The memory **128** is capable of storing an array of data described in detail below. The processing unit **124** is connected to the memory **128** and executes software instructions that are capable of being stored in a RAM of the memory **128** (e.g., during execution), a ROM of the memory **128** (e.g., on a generally permanent basis), or another non-transitory computer readable medium such as another memory or a disc. Software included in the implementation of the fastener driver **10** can be stored in the memory **128** of the controller **110**. The software includes, for example, an interrupt service routine (ISR), firmware, one or more applications, program data, filters, rules, one or more program modules, and other executable instructions. The controller **110** is configured to retrieve from the memory **128** and execute, among other things, instructions related to the control processes and methods described herein. In other constructions, the controller **110** includes additional, fewer, or different components.

[0025] During operation of the fastener driver **10**, a jam condition may occur from the drive blade

26 becoming stuck during its driving stroke somewhere between the TDC position and the BDC position. For example, a nail may buckle during a fastener driving cycle if it is unable to plunge into an underlying workpiece, jamming the buckled nail within the nosepiece of the driver **10** and preventing the driver blade **26** from reaching its BDC position. If the driver blade **26** is stopped between the TDC and BDC positions, the lifter rollers **90** may be unable to correctly mesh with the lift teeth **74**, resulting in a jam condition in which the lifter **66** and driver blade **26** remain seized. As described in more detail below, the jam detection system **100** is configured to adjust an operating condition of the fastener driver **10** when a jam condition occurs. During fastener driving events, it is assumed that if the piston **22** engages the bumper **98** at the BDC position the rollers **90** will correctly mesh with the lift teeth **74**. As such, as described above, jams typically occur in a fastener feeding area of the drive stroke (e.g., when the driver blade moves from the BDC position towards the TDC position).

[0026] Now with reference to FIG. 3, a jam condition may occur when the rollers **90** of the lifter **66** incorrectly mesh with the lift teeth **74**. In some embodiments, an unclearable jam condition may occur when one of the rollers **90** engages an end portion **74b** of one of the lift teeth **74** and is unable to roll to either side and lift the driver blade **26**. When the jam detection system **100** determines that an unclearable jam condition occurs, the indicator **112** is activated to alert the operator that maintenance is required.

[0027] The one or more sensors **101** (i.e., “driver blade sensors”) may include an optical sensor **102**, an inductive sensor **106**, a capacitive sensor **114**, a magnetic sensor **118**, and/or a lifter sensor **120**. One or more of the optical sensor **102**, the inductive sensor **106**, the capacitive sensor **114**, and magnetic sensor **118** are configured to detect a number of or each of the lift teeth **74** as the driver blade **26** is moved from the BDC position to the TDC position and as the driver blade **26** is released from the TDC position to the BDC position. The lifter sensor **120** that is configured to detect the angular position of the lifter **66**. The combination of the lifter sensor **120** and one or more of the optical sensor **102**, the inductive sensor **106**, the capacitive sensor **114**, and magnetic sensor **118** may be used in conjunction to detect whether a jam condition occurs.

[0028] As shown in FIG. 4, the fastener driver **10** includes a phototransistor **102a** (a type of optical sensor **102**) mounted to the lifter frame **70** on a first side of the driver blade **26** and an infrared emitter **102b** located on an opposite side of the driver blade **26**. Separately, the lifter sensor **120** is coupled to the lifter frame **70** and is configured to detect an angular position of the lifter **66**. As the driver blade **26** moves, a beam of light from the optical sensor **102** will be blocked when the lift teeth **74** intersect the beam. As such, the time that the beam is blocked can be used to calculate the width of the lift teeth **74** and time required for the driver blade **26** to move between predetermined positions (e.g., from the BDC position to the TDC position, the TDC position to the BDC position, and the BDC position to ready position). In other embodiments, the inductive sensor **106**, the capacitive sensor **114**, and/or magnetic sensor **118** may be used to detect the presence of the lift teeth **74** and determine a jam condition in a similar fashion as the optical sensor **102**.

[0029] To determine whether a jam condition occurs, the controller **110** is configured to timestamp edges of the teeth **74** as the teeth **74** block or pass the beam of light from the optical sensor **102**. When the driver blade **26** is released from the TDC position to the BDC position, the controller **110** determines whether a predetermined number of edges corresponding to the number of lift teeth **74** pass through the beam of light of the optical sensor **102**. In some embodiments, the controller **110** is configured to count the number of teeth **74** instead of counting the edges. If the number of detected edges are equal to the predetermined number of edges within a predetermined amount of time, the controller **110** determines that the driver blade **26** has reached the BDC position without a jam condition present. It should be appreciated that the predetermined number of edges may be the edges of all of the teeth **74** or a portion of the teeth (e.g., more than 50 percent of the teeth **74**). Additionally or alternatively, if the number of detected teeth are equal to the predetermined number of teeth within the predetermined amount of time, the controller **110** determines that the driver

blade **26** has reached the BDC position without a jam condition present.

[0030] For example, a fire-reload cycle (e.g., when the driver blade **26** is released from the TDC position to the BDC position and is returned towards the TDC position) without a jam condition is illustrated in FIG. **6**. During the cycle, the processing unit **124** executes a firmware where the lifter sensor **120** detects the position of the lifter **66** (e.g., illustrated as line **130**). Concurrently, the processing unit **124** is configured to activate a timer (e.g., illustrated as line **134**) and a counter (e.g., illustrated as line **138**) once the lifter sensor **120** detects that the lifter **66** moves through a nominal driver blade drop position, which is the position of the lifter **66** when the driver blade **26** is released (e.g., near the TDC position). The controller **110** counts the number of edges detected by the sensors **101** and timestamps the detection of each edge, which defines driver blade edge data. The controller **110** compares the number of detected edges to the predetermined number of edges corresponding to the number of lift teeth **74**. Once the controller **110** determines that the number of detected edges corresponds to the predetermined number of edges, the controller **110** activates a driver blade dropped flag. When the driver blade dropped flag is set prior to the predetermined amount of time, the controller **110** determines that a jam condition is not present and continues driving the motor **42** normally to complete the fire-reload cycle.

[0031] The controller **110** moves the driver blade edge data to an array in the memory **128** after the driver blade drop flag is activated. Therefore, the controller **110** is store the number of detected edges of the teeth **74** without processor intervention until the predetermined number of edges have been transferred. Concurrently, the controller **110** activates an interrupt service routine (ISR), which disables further transfer of driver teeth edge data to the processing unit **124** until the beginning of the next fire-reload cycle. In other words, the controller **110** disables counting the edges of the teeth **74** until a subsequent fire-reload cycle. This avoids logging potential extra edges when the driver blade **26** bounces after hitting the bumper **98**.

[0032] The lifter **66** begins to return the driver blade **26** from the BDC position towards the TDC position for a subsequent faster driving cycle. In some embodiments, the controller **110** may not monitor the movement of the driver blade **26** as the driver blade **26** moves to the TDC position. In other embodiments, the controller **110** may monitor movement of the driver blade **26** as the lifter **66** returns the driver blade **26** to a ready position, which is between the BDC position and the TDC position (e.g., if the operator does not actuate the trigger **58** for a subsequent driving cycle). In such an embodiment, the controller **110** may compare the number of detected edges of the teeth **74** to a second predetermined number of edges (e.g., corresponding to a number of lift teeth **74** less than the total number of lift teeth **74**). In other words, the controller **110** is configured to determine whether the driver blade **26** reaches the ready position within a predetermined amount of time. In the illustrated embodiment, the predetermined amount of time for a fire reload cycle may be in a range from 200 millisecond to 500 milliseconds while the predetermined time for the driver blade to move from the TDC position to the BDC position is 5 millisecond to 20 milliseconds

[0033] If, on the other hand, the controller **110** detects less than the predetermined number of edges within the predetermined amount of time while the driver blade **26** is moving toward the BDC position, the controller **110** determines that the driver blade **26** has not reached the BDC position and a jam condition is present. When a jam condition is present, the controller **110** does not activate the striker drop flag and applies a brake to the motor **42** to stop the motor **42** and lifter **66** from rotating. In addition, the controller **110** may activate the indicator **112** to alert the operator that a jam condition has occurred. In some embodiments, the indicator **112** may provide a first indication when the jam condition can be removed by the operator and a second indication when an unclearable jam condition occurs.

[0034] For example, a fire-reload cycle when a jam condition is present is illustrated in FIG. **7**. During the cycle, the processing unit **124** executes a firmware where the lifter sensor **120** detects the position of the lifter **66** (e.g., illustrated as line **140**). Concurrently, the processing unit **124** is configured to activate a timer (e.g., illustrated as line **144**) and a counter (e.g., illustrated as line

148) once the lifter sensor **120** detects that the lifter **66** moves through a nominal driver blade drop position. The controller **110** counts the number of edges detected by the sensors **101** and timestamps the detection of each edge. The controller **110** compares the number of detected edges to the predetermined number of edges corresponding to the number of lift teeth **74**. Once the controller **110** determines that the number of detected edges is less than or does not otherwise correspond with the predetermined number of edges, the controller **110** determines that there is a jam condition, and applies the brake to the motor **42** to stop the motor **42** and the lifter **66** from rotating. Moreover, the lifter **66** is stopped from further rotation to prevent any of the rollers **90** from jamming against or with the lifter teeth **74**, which might otherwise damage the drive blade **26**. [0035] FIGS. **8-12** illustrate another embodiment of a powered fastener driver **210**, with like components and features as the embodiment of the fastener driver **10** shown in FIGS. **1-7** being labeled with like reference numerals plus “**200**”. Only differences between the fastener driver **10** of FIGS. **1-7** and the fastener driver **210** of FIGS. **8-11** are specifically noted herein, such as a lifter position sensing assembly **295** that detects the position of a lifter **266**.

[0036] The gas spring-powered fastener driver **210** is operable to drive fasteners (e.g., nails, tacks, staples, etc.) held within a magazine **214** into a workpiece. The fastener driver **210** includes a cylinder **218** with a moveable piston **222** positioned within the cylinder **218**. The fastener driver **210** further includes a driver blade **226** that is attached to the piston **222** and moveable therewith. The fastener driver **210** includes a housing **230** having a cylinder housing portion **234** and a motor housing portion **238** extending therefrom. The cylinder housing portion **234** is configured to support the cylinder **218**, whereas the motor housing portion **238** is configured to support a drive unit **240**. The drive unit **240** includes an electric motor **242** and a transmission **282** positioned downstream of the motor **242**. In addition, the illustrated housing **230** includes a handle portion **246** extending from the cylinder housing portion **234**, and a battery attachment portion **250** coupled to an opposite end of the handle portion **246**. A battery pack **254** is removably coupled to the battery attachment portion **250** and supplies electrical power to the drive unit **240**. The handle portion **246** supports a trigger **258**, which is depressed by a user to initiate a driving cycle of the fastener driver **10**.

[0037] With reference to FIG. **9**, the driver blade **226** defines a driving axis **262** and includes a plurality of driver blade teeth or lift teeth **274** formed along an edge **278** of the driver blade **226**, which extends in the direction of the driving axis **262**. In particular, the lift teeth **274** project laterally from the edge **278** relative to the driving axis **262**. During a driving cycle, the driver blade **226** and piston **222** are moveable along the driving axis **262** between a top-dead-center (TDC) position and a bottom-dead-center (BDC) or driven position. The driver blade **226** may further be held in a ready position, which is positioned between the BDC position and the TDC position. The piston **222** is adjacent a top end **219** of the cylinder **218** in the TDC position, and the piston **222** is adjacent a bottom end **220** of the cylinder **218** in the BDC position.

[0038] The fastener driver **210** further includes a rotary lifter **266** supported within the housing **230** by a frame **270** (FIG. **10**). The rotary lifter **266** includes a plurality of pins **294** configured to engage the lift teeth **274** of the driver blade **226**. One or more of the pins **294** may include a roller **290** rotatably supported upon the pin **294**. The lifter **266** receives torque from the drive unit **240**, causing the lifter **266** to rotate. The lifter **266** and the drive unit **240** may be collectively referred to as a lifter assembly **288** (FIG. **8**). As the lifter **266** rotates, the pins **294** and the roller **290** sequentially engage the lift teeth **274** formed on the driver blade **226** to return the driver blade **226** along the driving axis **262** from the BDC position toward the TDC position.

[0039] The cylinder **218** includes a bumper **298** located at the bottom end **220** of the cylinder **218**. The bumper **298** has a generally annular, frusto-conical shape with a central bore **299** therethrough. The bore **299** is coaxial with the driving axis **262** such that the driver blade **226** extends through the bore **299**. As the piston **222** and the driver blade **226** move from the TDC position toward the BDC position, the piston **222** impacts the bumper **298**, which absorbs the impact from the piston **222** and

stops the piston **222** in the BDC position. In some embodiments, the bumper **298** is constructed of a resilient material (e.g., rubber, elastomeric material, or the like).

[0040] The fastener driver **210** includes an inductive sensor **275** (schematically illustrated in FIGS. **10** and **11**) coupled to the frame **270** by an inductive sensor housing **277**, a sensor target **285** coupled for co-rotation with the lifter **266**, and a controller **310** in communication with the inductive sensor **275**. For example, the controller **310** may include, among other things, a processing unit **324** (e.g., a microprocessor, a microcontroller, or another suitable programmable device) and a memory **328**. The sensor target **285** is configured to be sensed or detected by the inductive sensor **275**. It should be appreciated that the inductive sensor **275** may be coupled to any portion of the inductive sensor housing **277** so the sensor target **285** can be detected. The inductive sensor **275** and the sensor target **285** together define a lifter position sensing assembly **295** of a jam detection system **200** (FIG. **12**), which is described in more detail below. The lifter position sensing assembly **295** is configured to detect the angular position of the lifter **266** and the controller **310** is configured to stop the drive unit **240** in response to detecting that the angular position of the lifter **266** is abnormal during a fastener driving cycle.

[0041] The inductive sensor **275** is coupled to the frame **270** at a position below the lifter **266** by the inductive sensor housing **277**. In other words, the inductive sensor is positioned between the lifter **266** and the motor **242**. The sensor target **285** is separate from the lifter **266** and is made from a different material than the lifter **266** (e.g., for weight reduction). In other words, the lifter **266** is formed of a first material and the sensor target **285** is formed of a second material that is different than the first material (e.g., a lighter material). In other embodiments, the sensor target **285** may be integrally formed with the lifter **266** (e.g., a coating on the lifter **266** or the like). Further, in other embodiments, the inductive sensor **275** may be located above the lifter **266** and on an opposite side of the lifter **266** as the motor **242**. The combination of the inductive sensor **275** and the sensor target **285**, otherwise known as the lifter position sensing assembly **295**, allows the controller **310** to detect a position of the lifter **266** and may be used to detect abnormalities in the operation of the fastener driver **210**.

[0042] Now with reference FIG. **12**, the jam detection system **200** includes the lifter position sensing assembly **295**, the controller **310**, and an indicator **312** in communication with the controller **310**. The controller **310** includes a plurality of electrical and electronic components that provide power, operational control, and protection to the components and modules within the controller **310** and/or the fastener driver **210**. The indicator **312** may be exposed from an exterior of the housing **230** and may be configured as one or more lights (e.g., a light-emitting diode or LED), a display panel, or the like. The controller **310** may selectively activate the indicator **312** to alert the operator that maintenance is required.

[0043] During operation of the fastener driver **210**, a jam condition may occur from the driver blade **226** becoming stuck during its driving stroke somewhere between the TDC position and the BDC position. When the driver blade **226** is stuck, the lifter **266** becomes stuck as well, which results in an abnormal angular position of the lifter **266** during the fastener driving cycle. For example, a nail may buckle during a fastener driving cycle if it is unable to fully plunge into an underlying workpiece, jamming the buckled nail within the nosepiece of the driver **210** and preventing the driver blade **226** from reaching its BDC position. If the driver blade **226** is stopped between the TDC and BDC positions, the lifter pins **294** and/or rollers **290** may be unable to correctly mesh with the lift teeth **74**, resulting in a jam condition in which the lifter **266** and driver blade **226** remain seized.

[0044] The jam detection system **200** is configured to adjust an operating condition of the fastener driver **210** when a jam condition occurs. During a fastener driving cycle, it is assumed that the lifter **266** can rotate 360 degrees (e.g., a single complete revolution) in a predetermined amount of time. As such, the lifter position sensing assembly **295** communicates to the controller **310** to determine whether the lifter **266** rotates 360 degrees within the predetermined time. If not, the controller **310**

determines that a jam has occurred. In other words, the controller **310** is configured to stop the drive unit **240** in response to the lifter position sensing assembly **295** not detecting the angular position of the lifter **266** at the end of the fastener driving cycle. For example, the processing unit **324** may activate a timer in response to beginning of the fastener driving cycle (e.g., a user depressing the trigger **258** to activate the drive unit **240**). If the controller **310** does not receive an output from the inductive sensor **275** before expiration of the timer (e.g., indicating that the lifter has completed a full revolution), the controller **310** deactivates the drive unit **240**. When the jam detection system **200** determines that an unclearable jam condition has occurred, the indicator **312** is activated to alert the operator that maintenance is required.

[0045] It should be appreciated that the lifter position sensing assembly **295** described with reference to FIGS. **8-12** may be used in conjunction with the other sensors **101** described in FIGS. **1-7** (e.g., the lifter sensor **120**). As such, the description of the lifter sensor **120** applies equally to the lifter position sensing assembly **295**.

[0046] Various features of the disclosure are set forth in the following claims.

Claims

1. A powered fastener driver comprising: a housing; a cylinder within the housing, the cylinder containing a pressurized gas; a piston within the cylinder and moveable from a top-dead-center position to a bottom-dead-center position; a driver blade moveably coupled to the piston for driving a fastener into a workpiece, the driver blade having a plurality of teeth; a lifter configured to engage the plurality of teeth for returning the driver blade from the bottom-dead-center position toward the top-dead-center position; a drive unit configured to provide torque to the lifter, causing it to rotate; a sensor configured to detect one or more of the plurality of teeth; and a controller in communication with the drive unit and the sensor, the controller configured to: count a number of the plurality of teeth detected by the sensor, compare the number of detected teeth to a predetermined number of teeth, and stop the drive unit when the number of detected teeth is less than the predetermined number of teeth.
2. The powered fastener driver of claim 1, further comprising a lifter sensor configured to detect a position of the lifter, and wherein the controller is configured to activate a timer and a counter once the lifter sensor detects that the lifter moves through a nominal driver blade drop position.
3. The powered fastener driver of claim 2, wherein the controller is configured to execute an interrupt service routine when the number of detected teeth is equal to the predetermined number of teeth.
4. The powered fastener driver of claim 3, wherein the interrupt service routine disables the controller from counting the plurality of teeth until a subsequent fire-reload cycle.
5. The powered fastener driver of claim 2, wherein the controller is configured to determine whether the driver blade reaches a ready position within a predetermined amount of time.
6. The powered fastener driver of claim 1, wherein the controller is configured to determine whether the driver blade reaches the bottom-dead-center position within a predetermined amount of time.
7. The powered fastener driver of claim 6, wherein the controller includes a memory, and when the controller determines that the driver blade reached the bottom-dead-center position within the predetermined amount of time, the controller is configured to move data for the number of detected teeth to an array in the memory.
8. The powered fastener driver of claim 6, wherein the controller is configured to execute an interrupt service routine when the controller determines that the driver blade reached the bottom-dead-center position within the predetermined amount of time.
9. The powered fastener driver of claim 8, wherein the interrupt service routine disables the controller from counting the plurality of teeth until a subsequent fire-reload cycle.

- 10.** The powered fastener driver of claim 1, wherein the sensor is an optical sensor having a phototransistor positioned on a first side of the driver blade and an infrared emitter located on an opposite side of the driver blade.
- 11.** The powered fastener driver of claim 1, wherein the predetermined number of teeth are all the plurality of teeth or a portion of the plurality of teeth.
- 12.** A powered fastener driver comprising: a housing; a cylinder within the housing, the cylinder containing a pressurized gas; a piston within the cylinder and moveable from a top-dead-center position to a bottom-dead-center position; a driver blade moveably coupled to the piston for driving a fastener into a workpiece, the driver blade having a plurality of teeth; a frame supported within the housing; a lifter supported within the housing by the frame and configured to engage the plurality of teeth for returning the driver blade from the bottom-dead-center position toward the top-dead-center position; a drive unit configured to provide torque to the lifter, causing it to rotate; a lifter position sensing assembly configured to detect an angular position of the lifter at an end of a fastener driving cycle, the lifter position sensing assembly including a sensor coupled to the frame and a sensor target coupled for co-rotation with the lifter; and a controller in communication with the drive unit and the lifter position sensing assembly, the controller configured to stop the drive unit in response to the lifter position sensing assembly not detecting the angular position of the lifter at the end of the fastener driving cycle.
- 13.** The powered fastener driver of claim 12, wherein the lifter is formed of a first material and the sensor target is formed of a second material different than the first material.
- 14.** The powered fastener driver of claim 12, wherein the sensor target is separate from and coupled to the lifter.
- 15.** The powered fastener driver of claim 12, wherein the sensor target is integrally formed with the lifter.
- 16.** The powered fastener driver of claim 12, wherein the drive unit includes a motor, and wherein the sensor is positioned between the motor and the lifter.
- 17.** The powered fastener driver of claim 12, wherein the sensor is coupled to the frame by a sensor housing at a position below the lifter.
- 18.** The powered fastener driver of claim 12, further comprising a timer configured to be activated in response to a beginning controller of the fastener driving cycle, and wherein the controller is configured to deactivate the drive unit when the controller does not receive an output from the sensor before expiration of the timer.
- 19.** The powered fastener driver of claim 18, wherein the controller is configured determine whether the lifter rotated 360 degrees before the expiration of the timer.
- 20.** The powered fastener driver of claim 18, wherein the controller is configured to activate an indicator to alert an operator that maintenance is required when the controller does not receive the output from the sensor before expiration of the timer.
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